POWER PULSE

Power Generation (Hydro, Thermal, Gas, Nuclear, PSP, Solar, Wind)
Important Formulae

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1 Hydro-electric

1.

$$P = \rho g Q H \eta = 9.81\, Q\, H\, \eta \ \text{kW}$$

P = electrical power output (kW); ρ = density of water (1000 kg/m³); g = 9.81 m/s²; Q = water discharge (m³/s); H = net head of water, height of water level above the turbine less losses (m); η = overall efficiency (turbine × generator)

2.

$$E = P\,t, \qquad E_{\text{stored}}(\text{kWh}) = \frac{\rho g V H \eta}{3.6 \times 10^6} \approx \frac{V H \eta}{367}$$

E = energy (kWh); P = power (kW); t = time (h); V = volume of water stored (m³); H = head (m); η = overall efficiency; 3.6 × 10⁶ J = 1 kWh

3.

$$N_s = \frac{N\sqrt{P}}{H^{5/4}}$$

N_s = specific speed of the turbine; N = rotational speed (rpm); P = turbine power (kW); H = head (m). Pelton: low N_s (H > 300 m); Francis: medium (30–300 m); Kaplan: high N_s (H < 30 m)

4.

$$v_{jet} = C_v\sqrt{2gH}, \qquad u_{bucket} \approx 0.45\, v_{jet}$$

v_jet = velocity of the water jet (m/s); C_v = velocity coefficient of the nozzle (0.97–0.99); g = 9.81 m/s²; H = head at the nozzle (m); u_bucket = peripheral speed of the Pelton wheel buckets (m/s)

2 Thermal (Rankine cycle)

5.

$$\eta_{cycle} = \frac{(h_1 - h_2) - (h_4 - h_3)}{h_1 - h_4}\ ; \ \ \eta_{Carnot} = 1 - \frac{T_L}{T_H}$$

η_cycle = Rankine cycle efficiency; h = enthalpy of steam/water (kJ/kg): h_1 turbine inlet, h_2 turbine exit, h_3 condenser exit, h_4 pump exit; T_H, T_L = highest and lowest absolute temperatures of the cycle (K)

6.

$$\eta_{overall} = \eta_{boiler}\,\eta_{cycle}\,\eta_{turbine}\,\eta_{gen}\,\eta_{aux}$$

η_overall = station efficiency (heat in coal to electricity sent out); η_boiler = boiler efficiency; η_turbine = turbine efficiency; η_gen = generator efficiency; η_aux = auxiliary (station-use) factor

7.

$$\text{Heat rate} = \frac{860}{\eta_{overall}}\ \text{kcal/kWh}, \qquad \text{Coal(kg/h)} = \frac{P(\text{kW}) \times 860}{\eta_{overall}\, CV(\text{kcal/kg})}$$

Heat rate = heat needed per kWh; 860 kcal = heat equivalent of 1 kWh; P = station output (kW); CV = calorific value of coal (kcal/kg)

8.

$$\text{Steam rate} = \frac{3600}{h_1 - h_2}\ \text{kg/kWh}, \qquad \eta_{boiler} = \frac{\dot{m}_s(h_s - h_{fw})}{\dot{m}_f\, CV}$$

Steam rate = steam used per kWh; 3600 kJ = 1 kWh; ṁ_s = steam flow (kg/h); h_s = enthalpy of steam leaving boiler; h_fw = enthalpy of feed water; ṁ_f = fuel burnt (kg/h); CV = calorific value (kJ/kg)

3 Gas turbine (Brayton) and combined cycle

9.

$$\eta_{Brayton} = 1 - \frac{1}{r_p^{(\gamma-1)/\gamma}}, \qquad \frac{T_2}{T_1} = r_p^{(\gamma-1)/\gamma}$$

η_Brayton = ideal gas-turbine efficiency; r_p = pressure ratio of the compressor (p_2/p_1); γ = ratio of specific heats (1.4 for air); T_1 = compressor inlet, T_2 = compressor outlet temperature (K)

10.

$$W_{net} = c_p(T_3 - T_4) - c_p(T_2 - T_1), \qquad r_{p,opt} = \left(\frac{T_3}{T_1}\right)^{\frac{\gamma}{2(\gamma-1)}}$$

W_net = net work per kg of air (kJ/kg); c_p = specific heat of air at constant pressure (≈ 1.005 kJ/kg·K); T_3 = turbine inlet, T_4 = turbine outlet temperature (K); r_p,opt = pressure ratio for maximum net work

11. $$\text{Back-work ratio} = \frac{W_{compressor}}{W_{turbine}} \ (\approx 0.4\text{–}0.6), \qquad \eta_{cc} = \eta_g + (1 - \eta_g)\,\eta_s$$

W_compressor = work absorbed by the compressor; W_turbine = work produced by the turbine; η_cc = combined-cycle efficiency; η_g = gas-turbine efficiency; η_s = efficiency of the steam cycle fed by the exhaust heat

4 Nuclear

12. $$E = \Delta m\, c^2, \quad 1 \text{ amu} = 931 \text{ MeV}, \quad E_{fission} \approx 200 \text{ MeV}$$

E = energy released (J); Δm = mass lost, the mass defect (kg); c = speed of light (3 × 10⁸ m/s); amu = atomic mass unit; MeV = mega electron-volt (1 MeV = 1.602 × 10⁻¹³ J); E_fission = energy per fission of U-235

13. $$\text{fissions/s} = \frac{P}{200 \times 1.602 \times 10^{-13}} \approx 3.1 \times 10^{16} \text{ per MW}, \qquad \approx 1 \text{ g U-235 per MW-day}$$

fissions/s = number of U-235 nuclei splitting each second; P = thermal power (W); 200 × 1.602 × 10⁻¹³ J = energy of one fission; about 1 g of U-235 is fissioned per MW of thermal power per day

14. $$k = \frac{\text{neutrons in one generation}}{\text{neutrons in the previous generation}} = k_\infty P_{NL}, \quad k_\infty = \eta f p \varepsilon$$

k = multiplication factor (1 critical, > 1 supercritical, < 1 subcritical); k_∞ = factor for an infinite core; P_NL = non-leakage probability; η = neutrons produced per neutron absorbed in fuel; f = thermal utilisation factor; p = resonance escape probability; ε = fast fission factor

5 Pumped-storage plant (PSP)

15. $$P_{gen} = \rho g Q H\, \eta_t \eta_g, \qquad P_{pump} = \frac{\rho g Q H}{\eta_p \eta_m}$$

P_gen = output power in generating mode (W); P_pump = input power in pumping mode (W); ρ = 1000 kg/m³; g = 9.81 m/s²; Q = flow (m³/s); H = head between lower and upper reservoir (m); η_t = turbine, η_g = generator, η_p = pump, η_m = motor efficiency

16. $$\eta_{cycle} = \frac{E_{gen}}{E_{pump}} = \eta_p \eta_m\, \eta_t \eta_g \approx 0.70\text{–}0.80, \qquad E(\text{kWh}) \approx \frac{V H \eta}{367}$$

η_cycle = round-trip (cycle) efficiency; E_gen = energy generated (kWh); E_pump = energy used in pumping (kWh); V = volume of water moved to the upper reservoir (m³); H = head (m)

6 Solar photovoltaic

17. $$P = \eta\, A\, G\,; \quad P_{mp} = V_{mp} I_{mp}\,; \quad FF = \frac{V_{mp} I_{mp}}{V_{oc} I_{sc}}$$

P = module output (W); η = module efficiency; A = module area (m²); G = solar irradiance (W/m²; STC = 1000 W/m², 25 °C, AM 1.5); P_mp, V_mp, I_mp = power, voltage, current at the maximum power point; V_oc = open-circuit voltage; I_sc = short-circuit current; FF = fill factor

18. $$P(T) = P_{STC}\,[1 + \gamma\,(T_c - 25)], \qquad T_c = T_a + \frac{NOCT - 20}{800}\,G$$

P(T) = output at cell temperature T_c; P_STC = rated power at 25 °C; γ = temperature coefficient (≈ −0.004 per °C for silicon); T_c = cell temperature (°C); T_a = ambient temperature (°C); NOCT = nominal operating cell temperature (°C); G = irradiance (W/m²)

19. $$\text{Series: } V \text{ adds, same } I\,; \qquad \text{Parallel: } I \text{ adds, same } V\,; \qquad G_{tilt} = G_b \cos\theta + G_d + G_r$$

V = voltage, I = current of the modules; G_tilt = irradiance on the tilted panel (W/m²); G_b = beam (direct) component; θ = angle of incidence between the sun's rays and the panel normal; G_d = diffuse component; G_r = ground-reflected component

20. $$E = P_{kWp} \times PSH \times PR, \qquad CUF = \frac{E_{year}}{P_{rated} \times 8760}$$

E = daily energy (kWh); P_kWp = array rating at STC (kW peak); PSH = peak sun hours per day (h, equal to the daily insolation in kWh/m²); PR = performance ratio (0.75–0.85); CUF = capacity utilisation factor; E_year = annual energy (kWh); P_rated = rated power (kW); 8760 = hours in a year

21.

$$I = I_L - I_0\left(e^{qV/nkT} - 1\right), \qquad \delta = 23.45^\circ \sin\left[\tfrac{360}{365}(284 + n)\right]$$

I = cell current (A); I_L = light-generated current (A); I_0 = diode reverse saturation current (A); q = electron charge (1.602 × 10⁻¹⁹ C); V = cell voltage; n = diode ideality factor; k = Boltzmann constant; T = absolute temperature (K); δ = solar declination angle; n = day of the year (1 to 365)

7 Wind

22.

$$P_{wind} = \tfrac{1}{2}\rho A v^3\ ; \quad P_{out} = \tfrac{1}{2}\rho A v^3\, C_p\, \eta_m \eta_g\ ; \quad A = \pi R^2$$

P_wind = power in the wind (W); ρ = air density (1.225 kg/m³ at sea level, 15 °C); A = swept area of the rotor (m²); R = blade (rotor) radius (m); v = wind speed (m/s); C_p = power coefficient of the rotor; η_m = mechanical (gearbox) efficiency; η_g = generator efficiency

23.

$$C_{p,max} = \frac{16}{27} = 0.593\ (\text{Betz}), \qquad v_{rotor} = \frac{v_1 + v_2}{2} = \tfrac{2}{3}v_1,\ v_2 = \tfrac{1}{3}v_1$$

C_p,max = Betz limit, the maximum fraction of wind power that can be extracted; v_1 = wind speed upstream of the rotor (m/s); v_2 = wind speed far downstream (m/s); v_rotor = speed through the rotor plane (m/s)

24.

$$\lambda = \frac{\omega R}{v} = \frac{2\pi N R}{60\, v}, \qquad \frac{v_2}{v_1} = \left(\frac{h_2}{h_1}\right)^{\alpha}\ (\alpha \approx \tfrac{1}{7})$$

λ = tip-speed ratio (blade tip speed/wind speed); ω = angular speed of the rotor (rad/s); N = rotor speed (rpm); R = blade radius (m); v = wind speed (m/s); v_1, v_2 = wind speeds at heights h_1, h_2 (m); α = wind-shear exponent (≈ 1/7 for open country)

25.

$$CF = \frac{E_{year}}{P_{rated} \times 8760}, \qquad E_{year} = P_{rated} \times CF \times 8760$$

CF = capacity factor; E_year = energy generated in a year (kWh or MWh); P_rated = rated power of the turbine or farm; 8760 = hours in a year. Cut-in ≈ 3–4 m/s, rated ≈ 11–14 m/s, cut-out ≈ 25 m/s

8 Plant factors (all stations)

26.

$$\text{Capacity factor} = \frac{E_{actual}}{P_{rated} \times T}, \quad \text{Load factor} = \frac{P_{avg}}{P_{max}}, \quad \text{Plant use factor} = \frac{E_{actual}}{P_{rated} \times T_{operating}}$$

E_actual = energy generated in the period; P_rated = installed capacity; T = total hours in the period; P_avg = average load; P_max = maximum (peak) demand; T_operating = hours the plant was actually in service. Base load: nuclear, thermal, run-of-river; peak: hydro with storage, gas, PSP